Can a Cheap Text Monitor Tell Right from Wrong Reasoning?

CSI 436/536 Machine Learning (Fall 2026) — Project P Midterm One-Pager Group: Jiayu Liao, Zeyu Pan, Yuhan Sun Oct 20, 2026

Research question (fixed by the project).

Do pretrained representations improve downstream ML tasks, particularly under limited training data? We turn the yes/no question into a measurement problem. With the class algorithm g and training data fixed, the *only* varied factor is the feature source φ, and we report the gap function

$$\Delta(\theta) \;=\; L_{\text{test}}\big(g \circ \varphi_{\text{simple}};\, \theta\big) \;-\; L_{\text{test}}\big(g \circ \varphi_{\text{pretrained}};\, \theta\big), \qquad \theta = (\text{train size } n,\ \text{distribution shift},\ \text{perturbation}).$$

Deliverable: a measurement report — one summary sentence (yes/no with effect size and CI) plus three boundary curves $\Delta(n)$, Δ(shift), Δ(noise).

Task and data (application carrier).

A cheap *text monitor* for math reasoning: given a problem and one solution step (text only), classify the step as correct or incorrect. **PRM800K** (OpenAI, MIT license): human-labeled step-level correctness on MATH solutions (∼800k labels). **Math-Shepherd** (GSM8K, automatically labeled): transfer test set only. Perturbed variants (rule-based typos, number formatting, LaTeX style): robustness test sets. Labels come with the datasets; no LLM labeling is used.

Method (strictly within the rules).

Pretrained text embeddings — `all-MiniLM-L6-v2`, `bge-base`, `bge-large` — are used *frozen* as feature extractors only (forward pass, no fine-tuning). On top we train only models covered in class: logistic regression (L7) and linear SVM (L11). Control arm: TF-IDF (word / char n-grams) with the same classifiers. Protocol: input = problem + current step; **group split by problem** (no step of one problem crosses train/test, preventing leakage); primary metrics **balanced accuracy / AUROC** (labels are imbalanced; no raw accuracy); 10 stratified re-samples per n, mean ± CI; C selected by grouped k-fold CV inside the training set only; the test set is touched once; all seeds fixed.

Experiments.

E1 learning curves $\Delta(n)$, $n \in \{20, 50, 100, 500, 2000, \text{full}\}$ (sample efficiency); **E3** generalization Δ(shift): train on PRM800K (MATH) → test on Math-Shepherd (GSM8K) — reported as a *compound* shift (problem type, source, and labeling method change together; no single-factor attribution), plus train/test gap; **E4** robustness Δ(noise): AUROC degradation under rule-based perturbations; **E2** three embedding models (novelty axis); **E5/E6** optional (abstention via Chow's rule; a preliminary predictability probe).

Pre-registered predictions (locked Oct 19, before any runs).

(1) gap(n) peaks in the small-sample region ($n \leq 100$) and stays non-negative; (2) Δ(shift) remains positive; (3) embedding degradation under perturbation is no larger than TF-IDF's. Falsified predictions will be reported as boundary findings, not hidden.

Current progress.

Full pipeline implemented and smoke-tested (data prep → features → grouped CV → evaluation → figures; SST-2 bring-up passed). A first run on PRM800K exposed a *label-construction bug on our side*: using only the `chosen` completion per step yielded a 99.8% positive label rate and degenerate metrics. We fixed it to use all human-rated completions and are re-running; all current numbers are preliminary. Encouragingly, the ranking-based signal was unaffected by the bug: at small n the embedding–TF-IDF gap is consistently positive (e.g., +0.03 AUROC at n=50, MiniLM, preliminary).

Challenges and plans.

Class imbalance (stratified sampling + balanced metrics); compound-shift attribution; honest reporting if pre-registration is falsified. Next: finish E1 (top priority) → E3 → E4 → E2 on the corrected dataset; final talk will show $\Delta(n)$, Δ(shift), Δ(noise).